

WASP-43b

R-2256

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_150304 · created 2026-08-16T00:41:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457085.85875 +/- 0.00089 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.012 +/- 0.055
Transit depth (Rp/Rs)^2	0.015 +/- 0.14 [%]
Orbital Inclination (inc)	77.63 +/- 2.95
Airmass coefficient 1 (a1)	2.02 +/- 0.8
Airmass coefficient 2 (a2)	-0.4 +/- 0.35
Scatter in the residuals of the lightcurve fit is	45.26 %
Best Comparison Star	#1 - [112, 47]
Optimal Aperture	1.43
Optimal Annulus	8.25
Transit Duration (day)	0.015 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.012 ± 0.055 vs 0.1594 (deviation 2.5σ)
Duration fit vs published	0.015 d vs 0.0512 d (deviation 1.99σ)
Mid-time O–C	0.9 min
Depth SNR	0.2
Residual scatter	45.26 %

Light curve

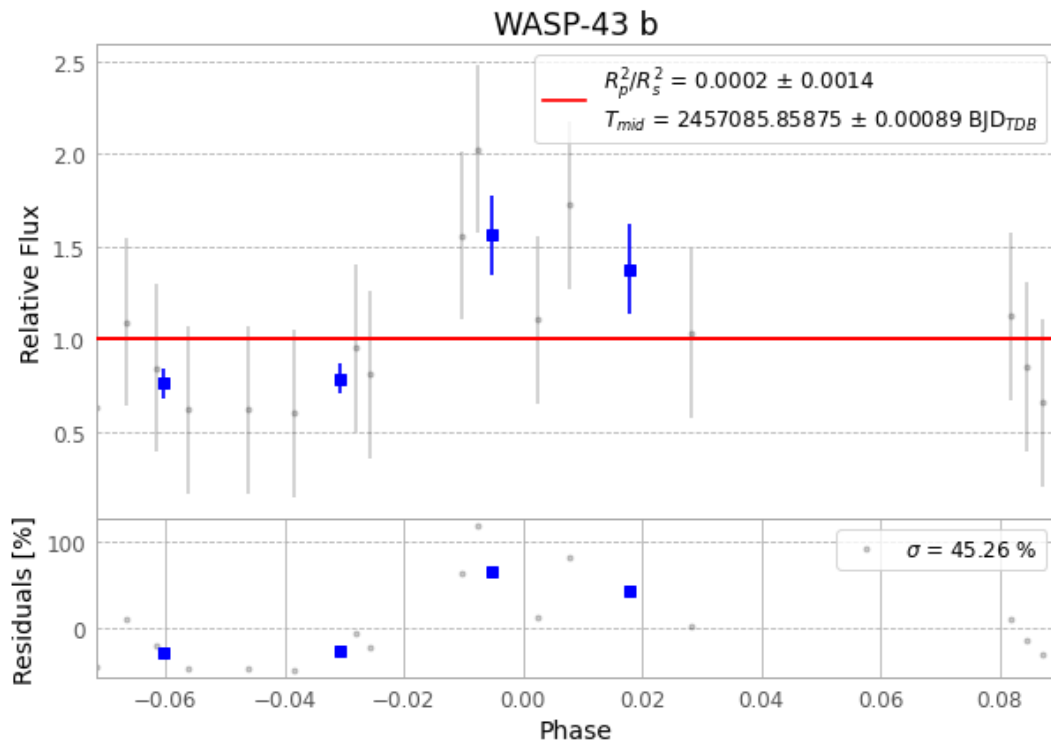


Figure 1 — FinalLightCurve_WASP-43 b_2015-03-04.png (SHA-256 4dfe7dafcbd03e48...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c6fb85a83c66be57...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), WASP-43150304070312.FITS → WASP-43150304101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-150304.log (2944aea0a4e7...), FinalLightCurve_WASP-43 b_2015-03-04.png (4dfe7dafcbd0...), FinalLightCurve_WASP-43 b_2015-03-04.csv (67322b2fa2ec...)

Compute resources

Wall time 174.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:41Z.